

ECEN-743: Homework 1

Due on February 8, 2026 at 6:00 PM

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Problem 1

Let $x, y \in \mathbb{R}^n$. The triangle inequality states that $\|x + y\| \leq \|x\| + \|y\|$. Use this to show that $\|x - y\| \geq \|x\| - \|y\|$.

Solution

Problem 2**Solution**

Problem 3**Solution**

Problem 4**Solution**

Problem 5**Solution**

Problem 6**Solution**

Problem 7**Solution**